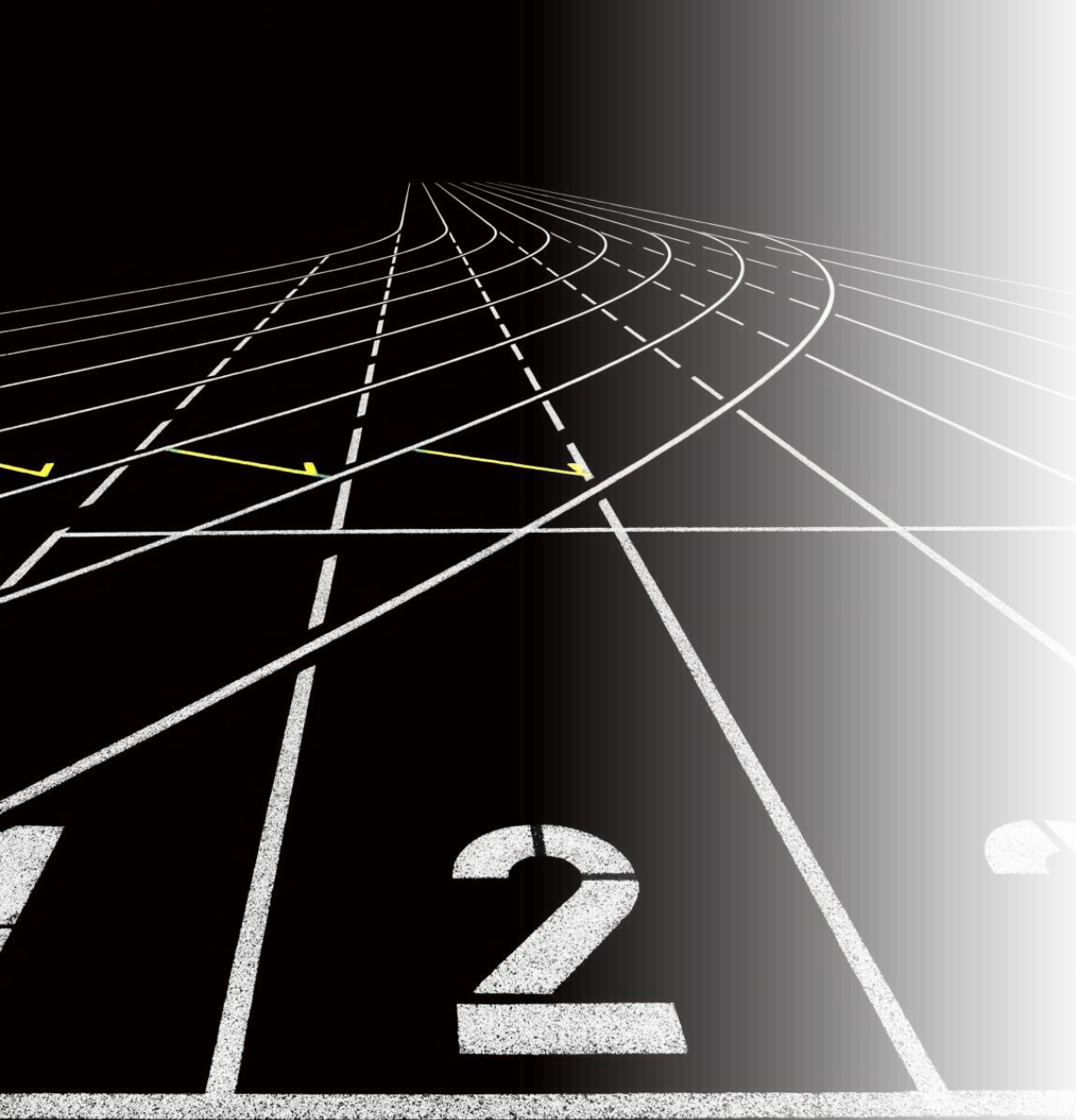


MECHANICS





Description of Motion

Chapter 14 (Part1)



Mechanics (Grade 10)

List of Terms

- Relative Motion
- Space Frame of Reference
- Coordinates System
- Position

- Displacement
- Trajectory
- Time Reference
- Duration

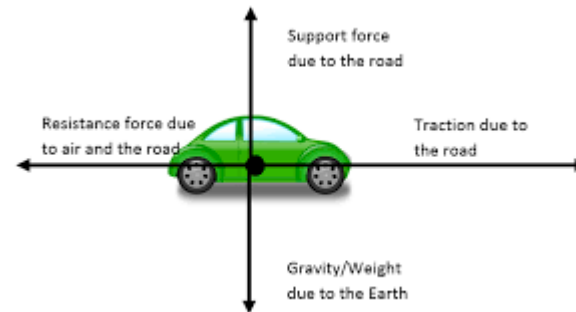
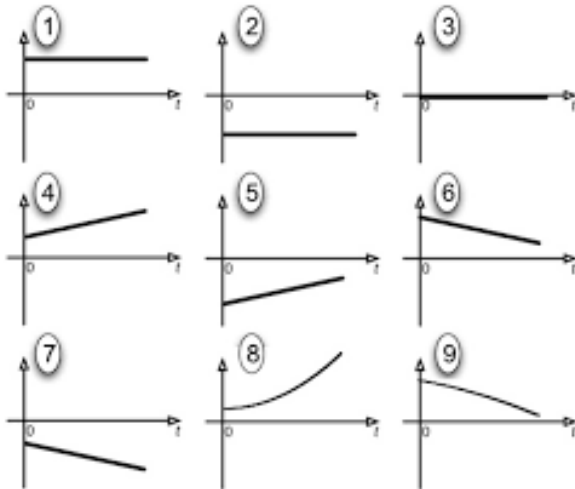
What is Mechanics?

Mechanics is the **branch of physics** that studies **motion** of objects.

Mechanics

Kinematics

Dynamics



Kinematics

Kinematics is the branch of mechanics that studies or describes the motion of a moving body **without being interested in the causes of its movement.**

Mechanics

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graph LR; Mechanics[Mechanics] --> Kinematics[Kinematics]; Mechanics --> Dynamics[Dynamics]; Kinematics --> K1[Just describes how is the object moving]; Kinematics --> K2[Neglects forces]; Kinematics --> K3[Uses words like: time interval, speed, velocity, acceleration, ...]; Dynamics --> D1[Studies effect of forces on motion]; Dynamics --> D2[Uses words like: weight, friction, normal reaction, tension, acceleration, ...];
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Kinematics

Just describes
how is the object moving

Neglects forces

Uses words like: time
interval, speed, velocity,
acceleration, ...

Dynamics

Studies effect of forces on
motion

Uses words like: weight,
friction, normal reaction,
tension, acceleration, ...

Mechanics (Grade 10)

List of Terms

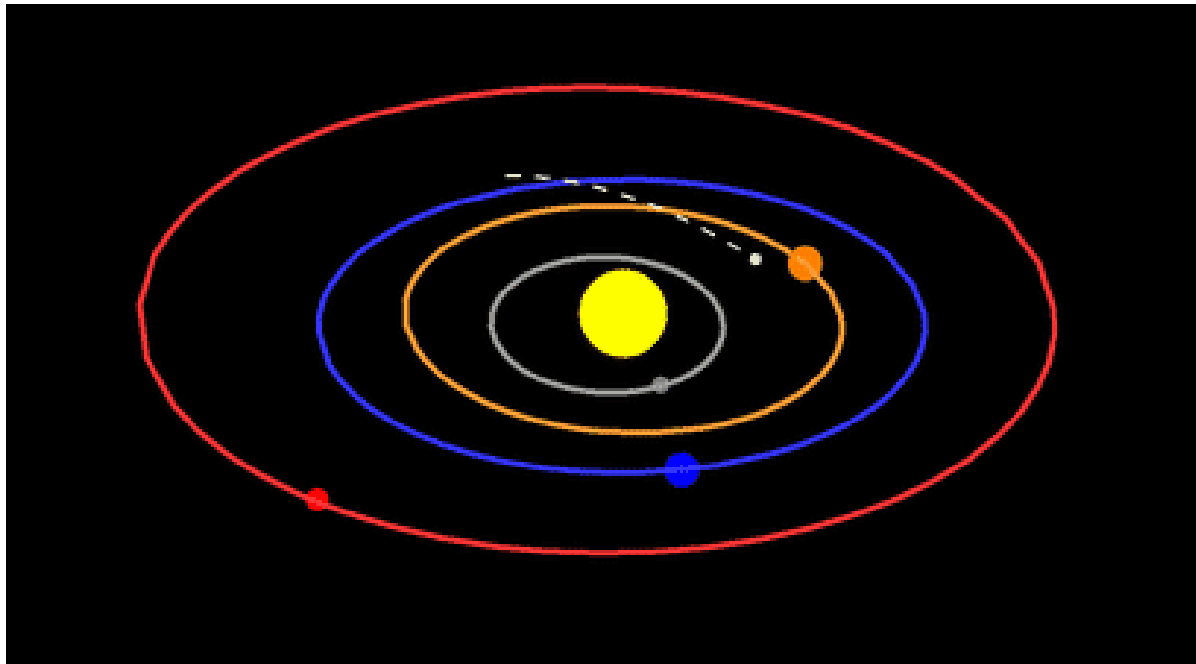
- Relative Motion
- Space Frame of Reference
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- Position

- Displacement
- Trajectory
- Time Reference
- Duration

Why do we need to choose a
Space Frame of Reference?

Rest or Motion?

Is the whiteboard in our classroom moving or at rest? Why?



Conclusion

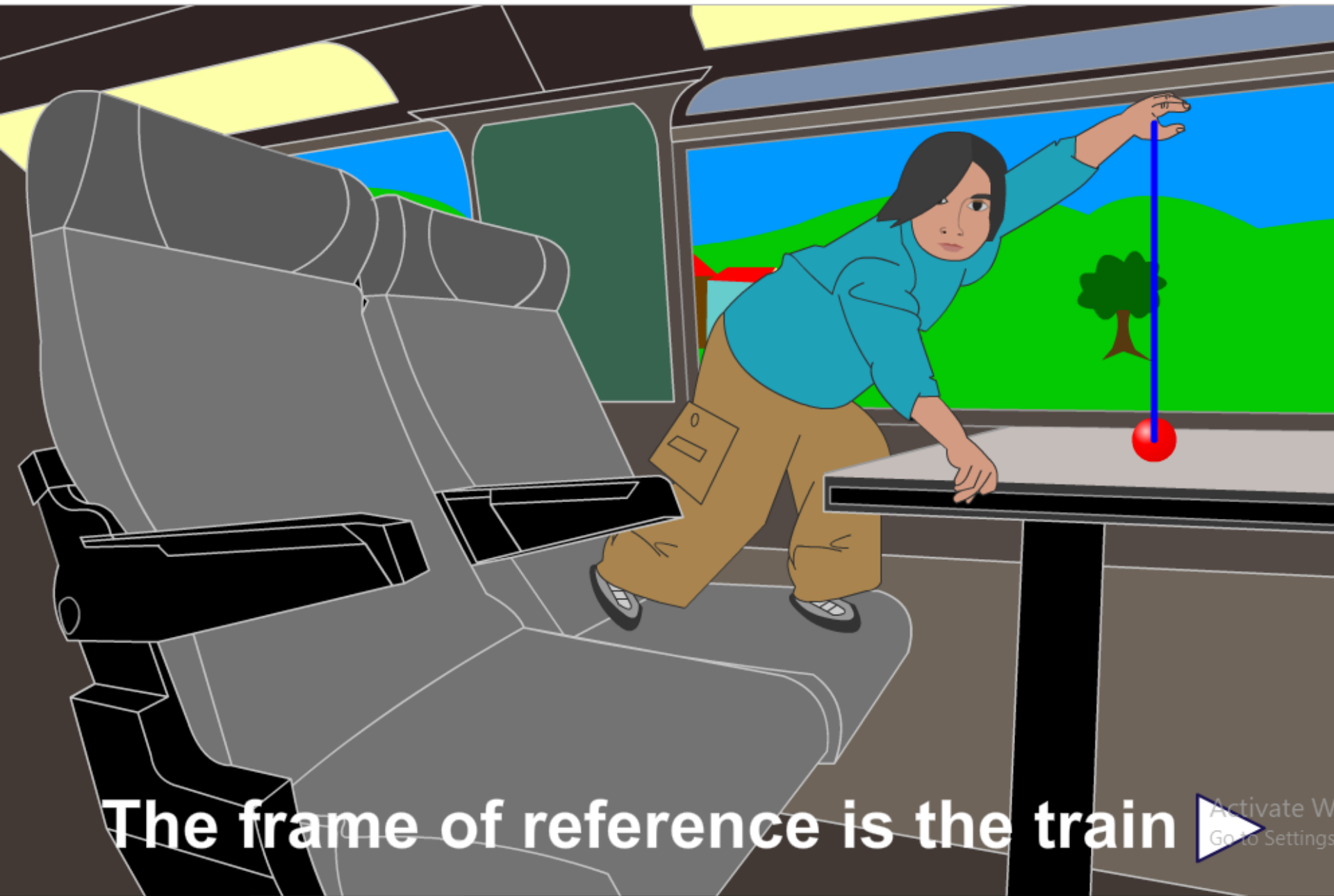
The concepts of **rest** and **motion** are **relative**: the same object may be at rest relative **to an observer** and moving relative **to another**.

Space Frame of Reference

- Since the movement has a **relative character**, the same object can be at the same time at rest with respect to one observer but movable with respect to another.
- Therefore, motion must be studied in a **given frame of reference**.
- The **description** of the movement **is not the same** in different frames of reference.
- By **default**, if it is not specified, the terrestrial (**earth**) frame of reference is chosen.

Motion Relativity - Another Example

Concerning a ball dropped by a person sitting in a train, compare its motion from the point of view of an observer sitting in the **train** (train is reference) with an observer attached to the **ground** (earth is reference).



The frame of reference is the train

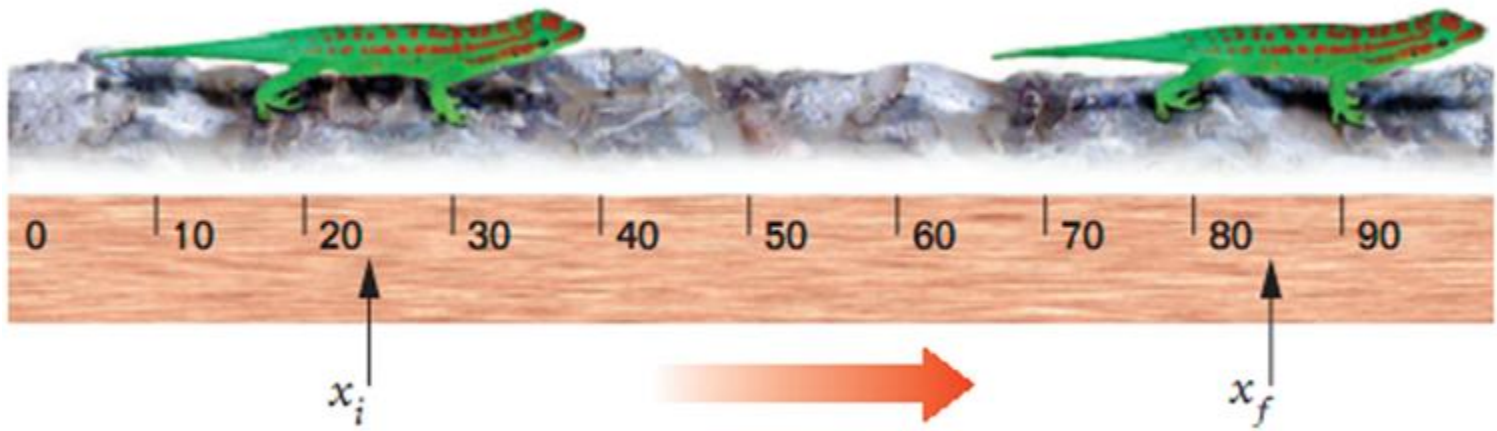
Activate W
Go to Settings



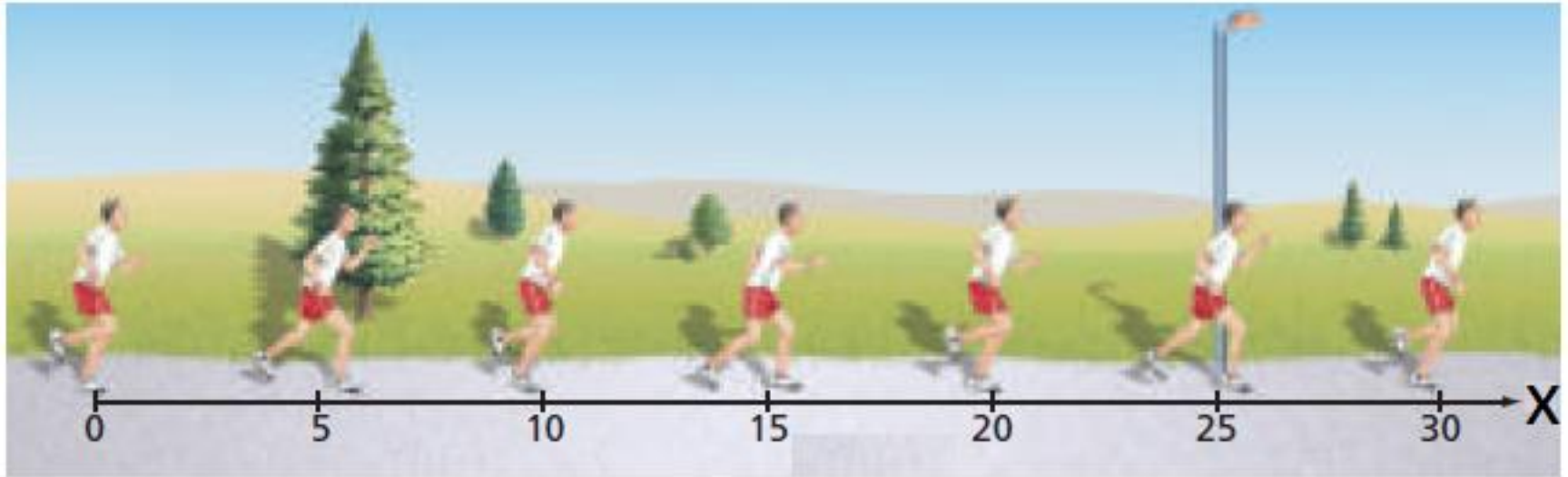
The frame of reference is the Earth

Activate Windows
Go to Settings to activate

Locating an Object - Coordinate System

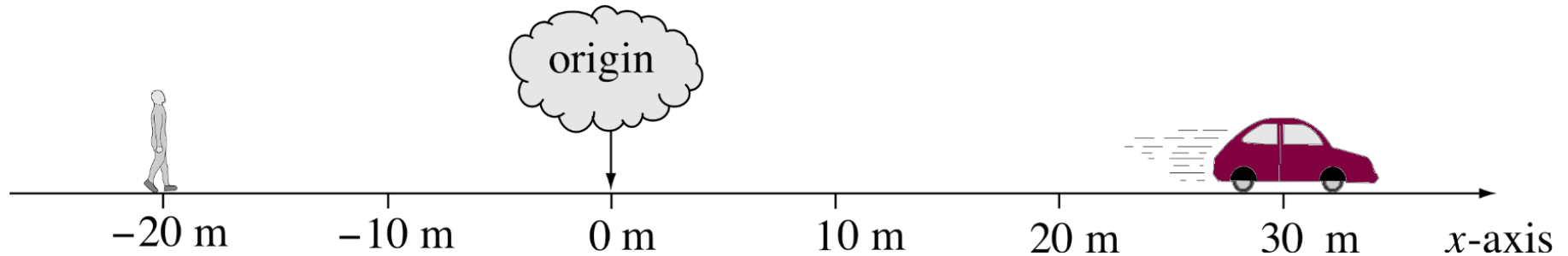


Coordinate System – Origin of Space



- The x-axis, oriented in the direction of the motion, is called coordinate System.
- The starting position, labelled O, can be chosen as the origin of space.
- The position M of the mobile, at a time t, can be told on this axis by the abscissa : $x = \overline{OM}$.

Evaluation



What is the abscissa for each of the person and the car shown in the figure?

Time Reference



- For example, the watch on your room wall (local time) can be used as a time reference.
- It tells you the date of events.

Duration

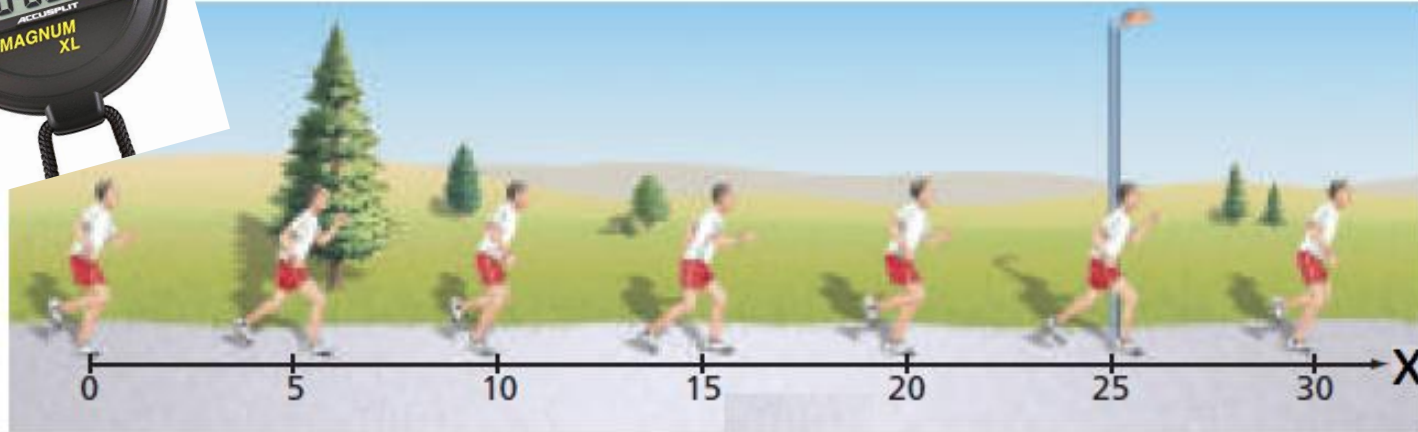


- The time between 2 instants or two given dates is called duration(Δt).

$$\Delta t = t_f - t_i$$

- Δt is measured in SI units in seconds (s).

Origin of Time

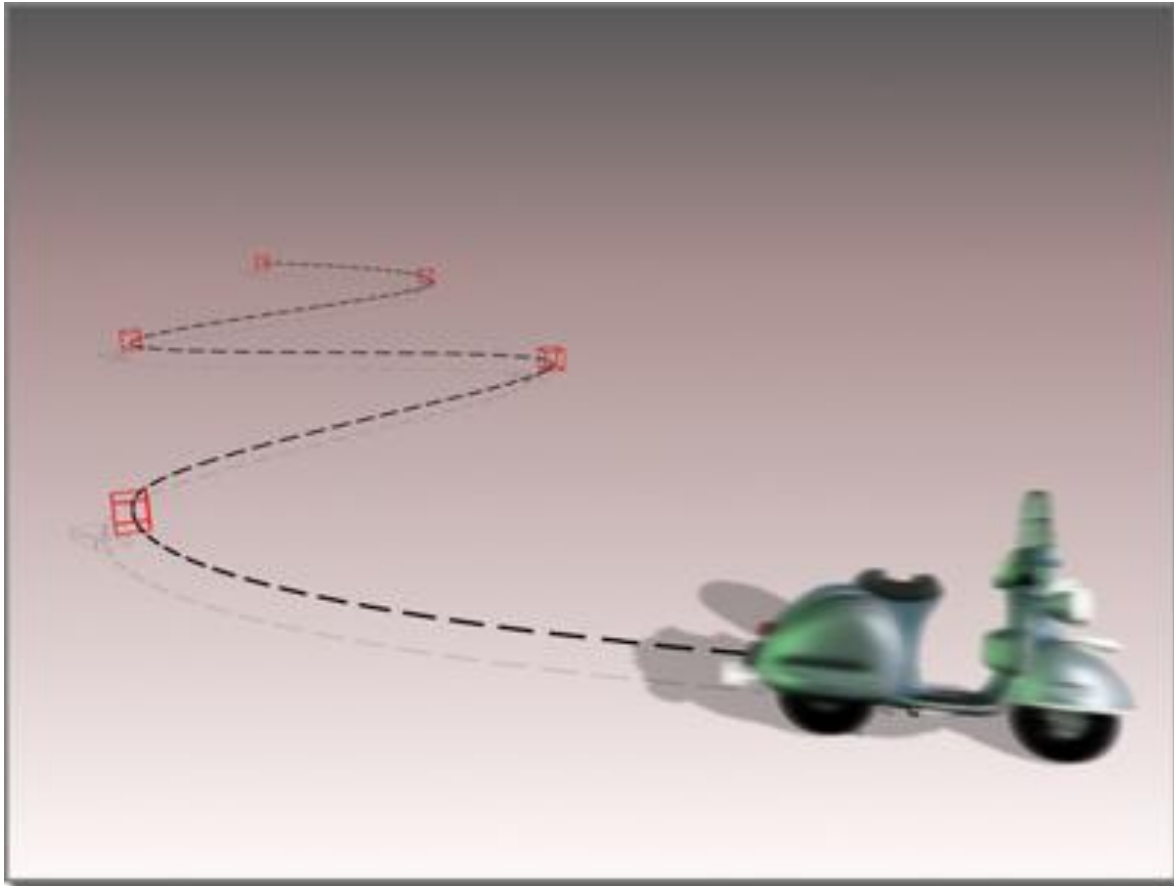


- We can take the **moment of departure** as the **origin of time**.
- A stopwatch may be used for counting duration.

Trajectory



Trajectory



Each point of this trajectory represents the position of the mobile at each instant of time.

Trajectory

This is the path followed by an object while moving.

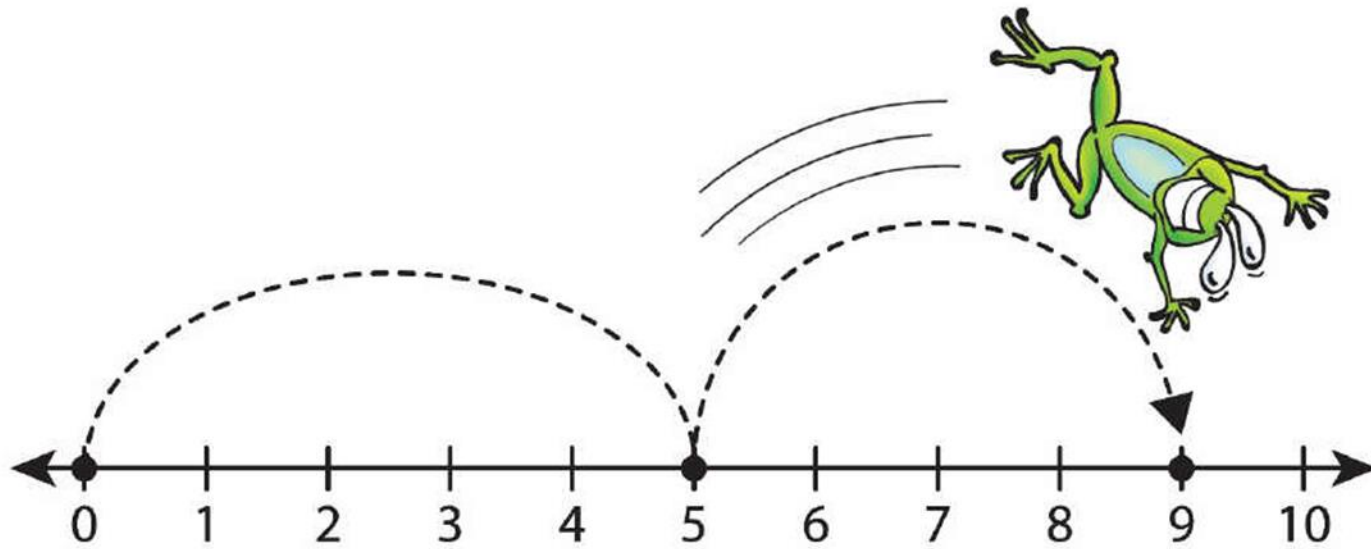
Rectilinear Trajectory



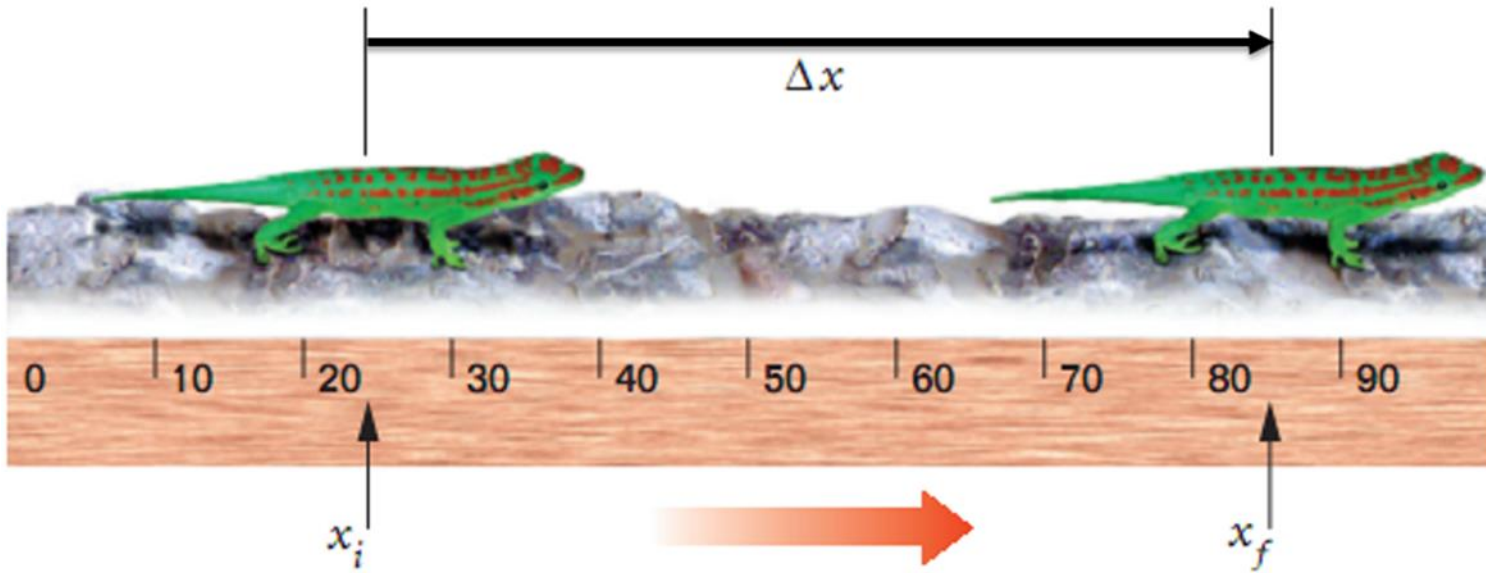
Curvilinear Trajectory



Displacement



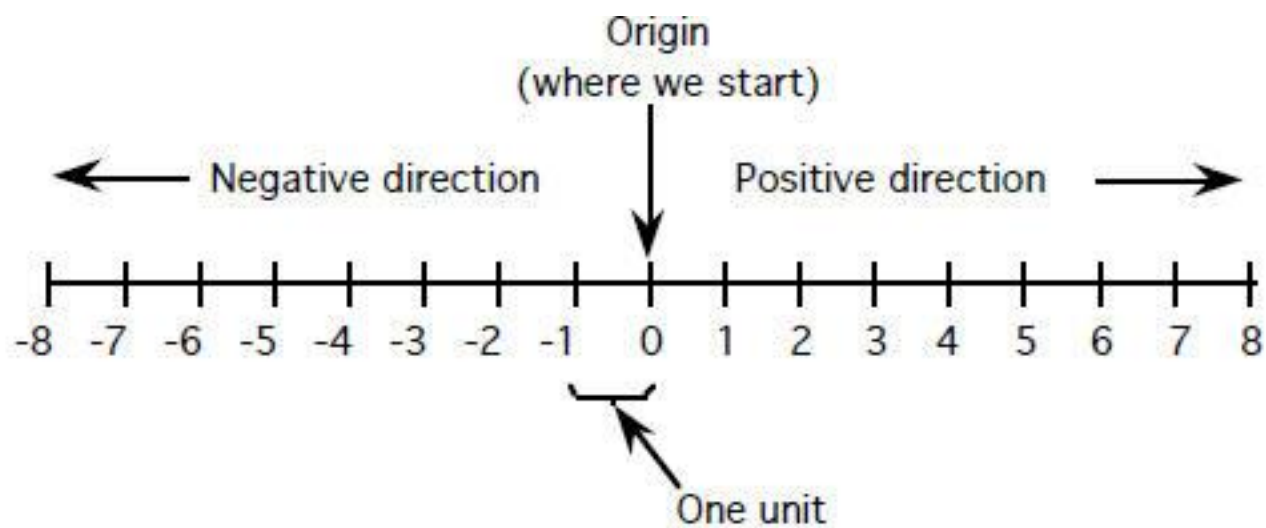
Displacement



- The change in location or place between 2 locations is called displacement (Δx).

$$\Delta x = x_f - x_i$$

- Δx is measured in SI units in meters (m).



Evaluation

We saw that :

- Motion and rest are two relative concepts.
- The description of motion of an object depends on the frame of reference chosen.
- The line or path described by an object during its motion is called trajectory.
- When the trajectory is a straight line, the motion is rectilinear.
- Usually, on an axis, the starting point is chosen as origin of space and the starting time as origin of time.
- The position of a point M on an axis is determined by its abscissa x .
- The time interval separating two instants is called the duration.

Summary

- ✓ The concepts of rest and motion are relative, and the description of motion depends on the frame of reference chosen.
- ✓ Trajectory is the path followed by a moving object.
- ✓ On $\overrightarrow{x'Ox}$ the position of a mobile M can be determined by its abscissa:

$$x_M = \overline{OM}$$

- ✓ The origin of space is characterized by: $x_O = 0$
- ✓ Displacement (or change in position) from M_1 to M_2 is found by:

$$\Delta x = x_f - x_i$$

- ✓ The time interval between 2 instants or two given dates is called duration(Δt):

$$\Delta t = t_f - t_i$$

- ✓ The starting moment is *normally* considered as the origin of time ($t_i = 0$).